

ROOT CAUSE HYPOTHESIS DOCUMENT

Root Cause Analysis of Defective Batches in Food Processing

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OBJECTIVE

To identify underlying reasons for batch quality deviations by analyzing patterns across shifts, materials, processes, and sensors, enabling proactive defect mitigation.

DEFECT FREQUENCY & DISTRIBUTION

OBSERVATIONS:

- Defect rates vary significantly across different product codes.
- Some defect types (e.g., overheating-related) recur in specific batches or lines.
- Certain suppliers and material grades are associated with consistently higher defect rates.

HYPOTHESES:

1. Defect frequency is influenced by material quality or inconsistencies in supplier inputs.

SHIFT AND OPERATOR PATTERNS

OBSERVATIONS:

- Higher defect percentages occur during Night Shifts or certain Shift_IDs.
- Operator-linked summaries reveal that a few operators consistently oversee defective batches.
- The intersection of Shift_Type and Test_Type (from QC logs) shows some combinations underperform.

HYPOTHESES:

1. Operational lapses are more frequent during low-staffed or less supervised shifts.

PROCESS & SENSOR-LEVEL ISSUES

OBSERVATIONS:

- Overheating events are higher on specific lines, correlating with higher defect rates.
- Sensor logs show temperature and pressure anomalies for batches flagged as defective.
- Box plots reveal outliers in sensor readings linked to poor quality output.

HYPOTHESES:

1. Inconsistent machine conditions (e.g., temperature spikes, pressure dips) are causing quality issues.

MATERIAL TRACEABILITY

OBSERVATIONS:

- Defect patterns align with specific material types, grades, and supplier IDs.
- Material with low Quality_Grade scores tends to cluster around defective batches.

HYPOTHESES:

1. Traceability gaps in raw material quality are leading to undetected risks upstream.

NEXT STEPS:

1. Validate hypotheses using cross-filtered charts and drill-throughs in Power BI.
2. Prioritize issues with highest frequency and severity.
3. Propose targeted process improvements (e.g., sensor alerting, shift retraining, supplier reviews).